Digital tech Summit 2026

BikeNetKit: Open Bicycle Network Algorithms for boosting the Green Transition of European Cities

Michael Szell



BikeNetKit

NERDS.itu.dk

IT UNIVERSITY OF COPENHAGEN

Nov 4, 2026

THE SETTING Cities have inefficient, car-centric transport

It comes with many issues







THE SETTING Cities have inefficient, car-centric transport

It comes with many issues







Each km travelled by Car costs society 0.11 €



Cities bear massive public health costs!

-500 Billion EUR
per year in the EU

Gössling et al, Ecol Econ 158 (2019)
Fosgerau et al, PNAS 120.16 (2023)

THE POTENTIAL Cycling is cost-efficient and time-tested

It would solve the issues



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THE POTENTIAL Cycling is cost-efficient and time-tested

It would solve the issues



Each km travelled by Bike brings society 0.18 €



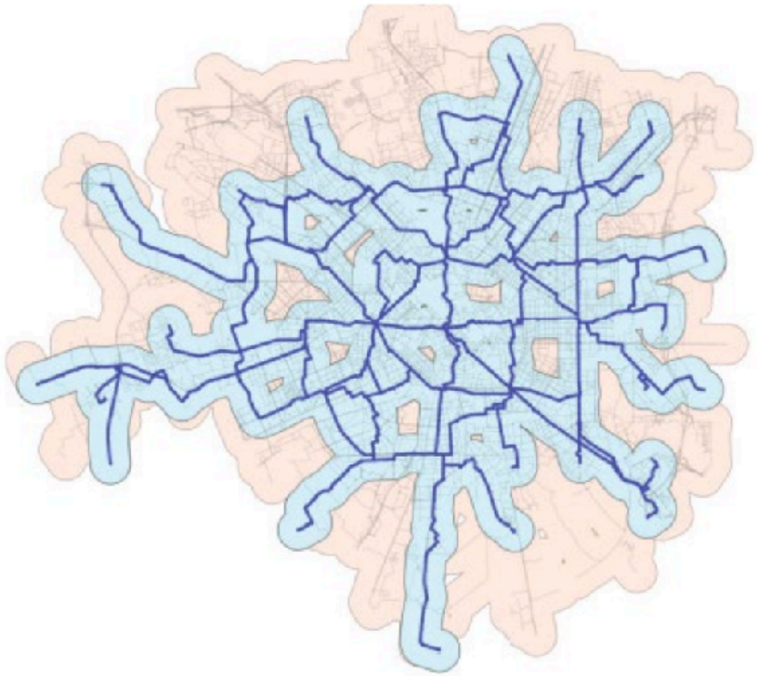
Cities gain massive public health benefits + hit climate targets!

+24 Billion EUR
per year in the EU
(currently)



Gössling et al, Ecol Econ 158 (2019)
Fosgerau et al, PNAS 120.16 (2023)

THE POTENTIAL Cycling is cost-efficient and time-tested



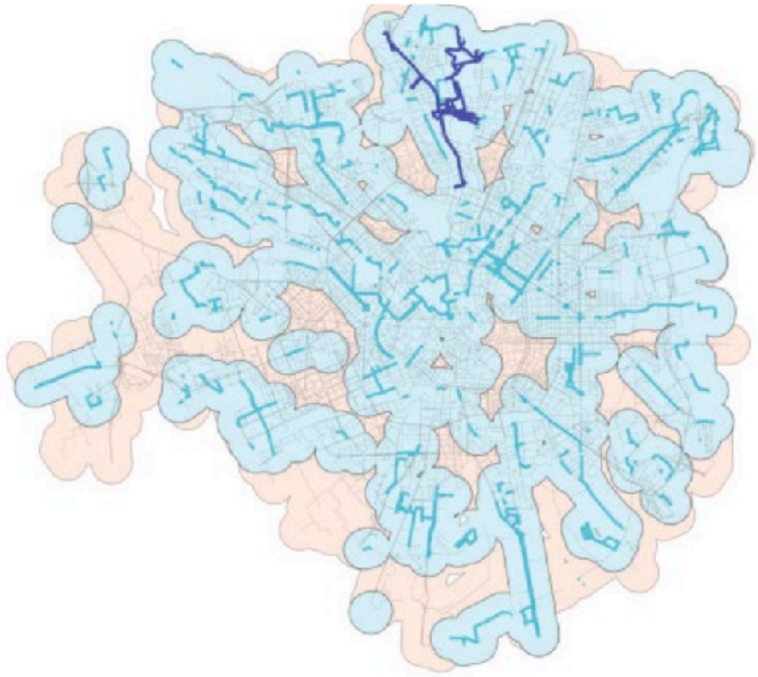
185km

>90 Mio.€

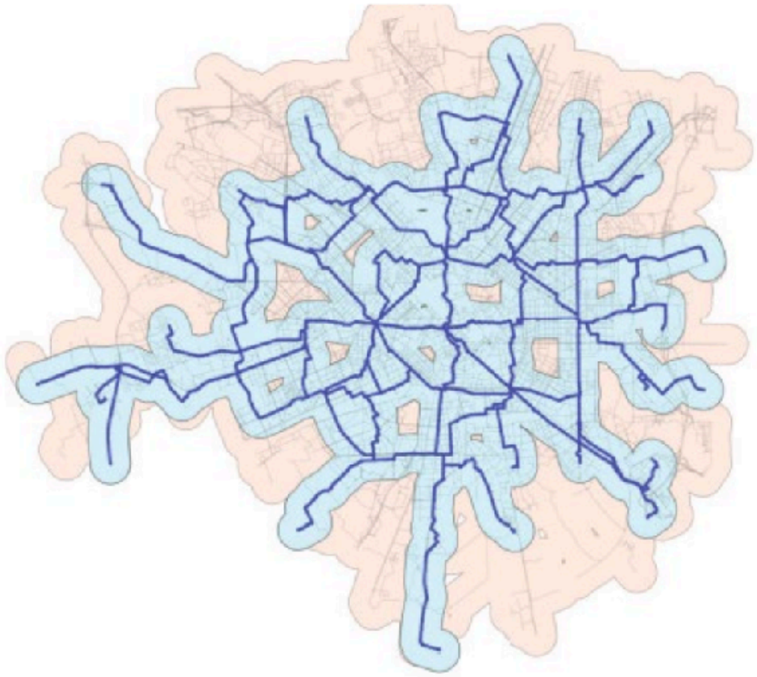
1 km of bike lane:
Breaks even after 3y
Then, +**430000**€ / y!

Szell et al, Sci Rep 12 (2022)
Fosgerau et al, PNAS 120.16 (2023)

THE PROBLEM Cities struggle with bike network planning



Real Milan



Synthetic Milan

185km

>90 Mio.€

Our research shows:
Data-driven planning is highly cost-efficient!

Szell et al, Sci Rep 12 (2022)

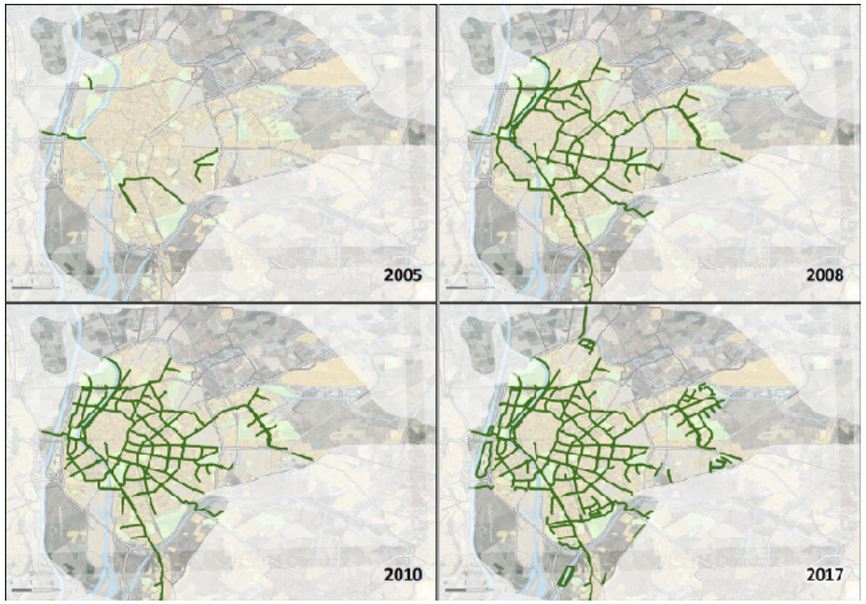
THE PROBLEM Cities struggle with bike network planning

Current network planning is:
- manual
- political
- lacks success metrics

Where to close the gaps?



How to build from scratch?

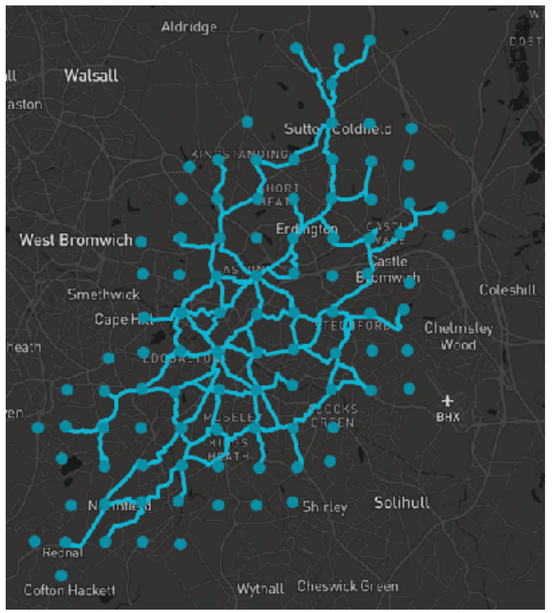


OUR APPROACH Bicycle network algorithms, supporting planners

Here to close the gaps!



This way to build from scratch!



OUR APPROACH Bicycle network algorithms, supporting planners

We have pioneered a ***Science of Bicycle Networks***

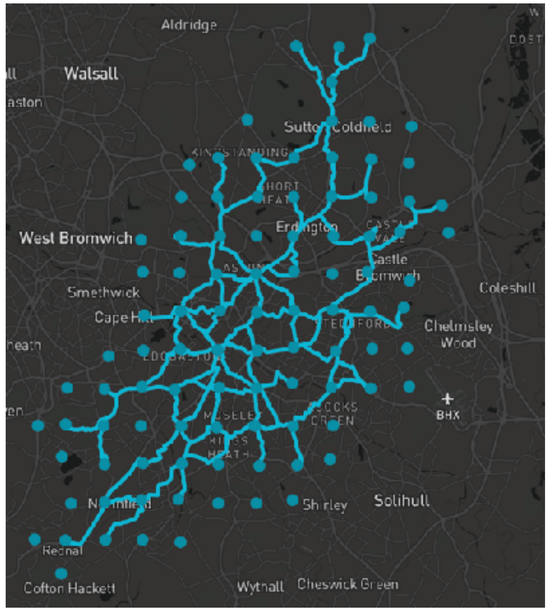
Key research/algorithms

Sebastiao et al, Appl Net Sci (2026)
Szell et al, arxiv:2604.07029 (2026)
Vybornova et al, EPB 52 (2025)
Lonardi et al, R Soc Int 22 (2025)
Vierø et al, EPB 51 (2024)
Büth et al, JOSS 9 (2024)
Folco et al, EPB 50 (2023)
Vybornova et al, Geog Anal 55 (2023)
Szell et al, Sci Rep 12 (2022)
Breum et al, Transp F 56683 (2022)
Natera et al, R Soc Open Sci 7 (2020)

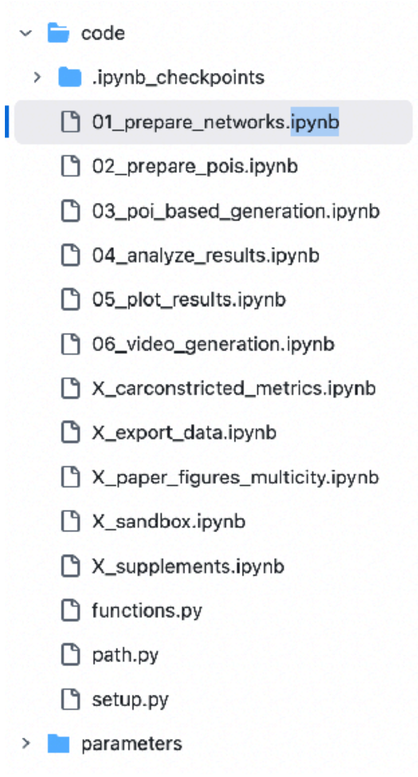
Here to close the gaps!



This way to build from scratch!



OUR PROBLEM Our algorithms were not user-friendly



Download and wrangle data

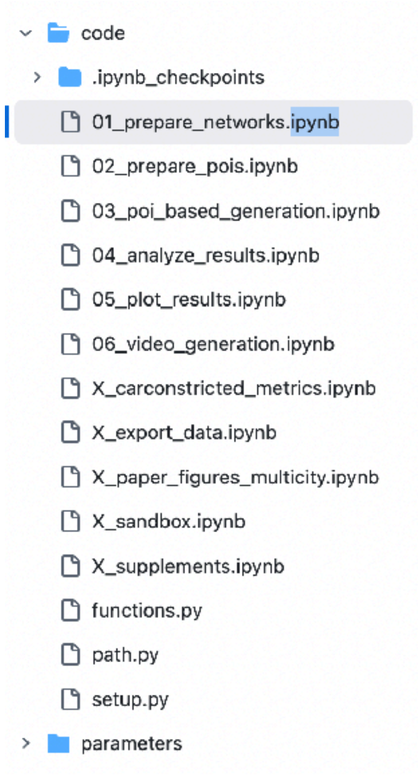
Networks

In [ ]:

```
for placeid, placeinfo in tqdm(cities.items(), desc = "Cities"):
    if placeinfo["nominatimstring"] != '':
        location = ox.geocoder.geocode_to_gdf(placeinfo["nominatimstring"])
        location = fill_holes(extract_relevant_polygon(placeid, shapely.geometry.shape(location['geometry'][0])))
        if debug: # Draw location polygons and their holes
            try:
                color = cm.rainbow(np.linspace(0,1,len(location)))
                for poly,c in zip(location, color):
                    plt.plot(*poly.exterior.xy, c = c)
                    for intr in poly.interiors:
                        plt.plot(*intr.xy, c = "red")
            except:
                plt.plot(*location.exterior.xy)
            plt.show()
    else:
        # https://gis.stackexchange.com/questions/113799/how-to-read-a-shapefile-in-python
        shp = fiona.open(PATH["data"] + placeid + "/" + placeid + ".shp")
        first = next(iter(shp))
        try:
            location = Polygon(shapely.geometry.shape(first['geometry'])) # If shape file is given as linestring
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    Gs = {}
    for parameterid, parameterinfo in tqdm(osmnxparameters.items(), desc = "Networks", leave = False):
        for i in range(0,10): # retry
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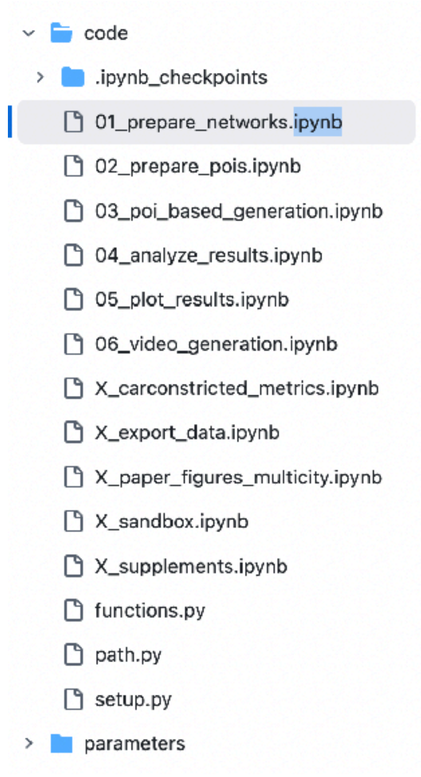
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Jupyter notebooks

Spaghetti code



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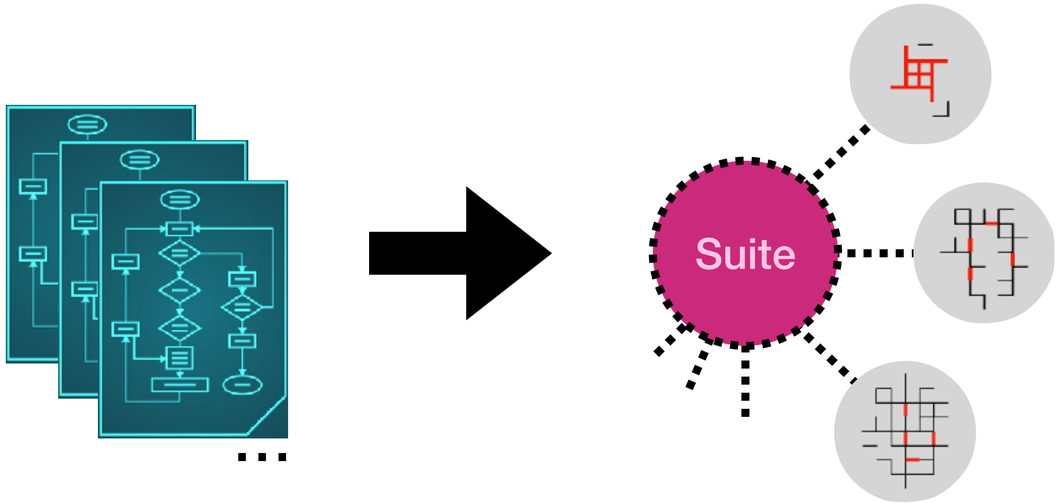
Jupyter notebooks

Spaghetti code



We got funds to 1) Create usable software

Innovation Fund Denmark

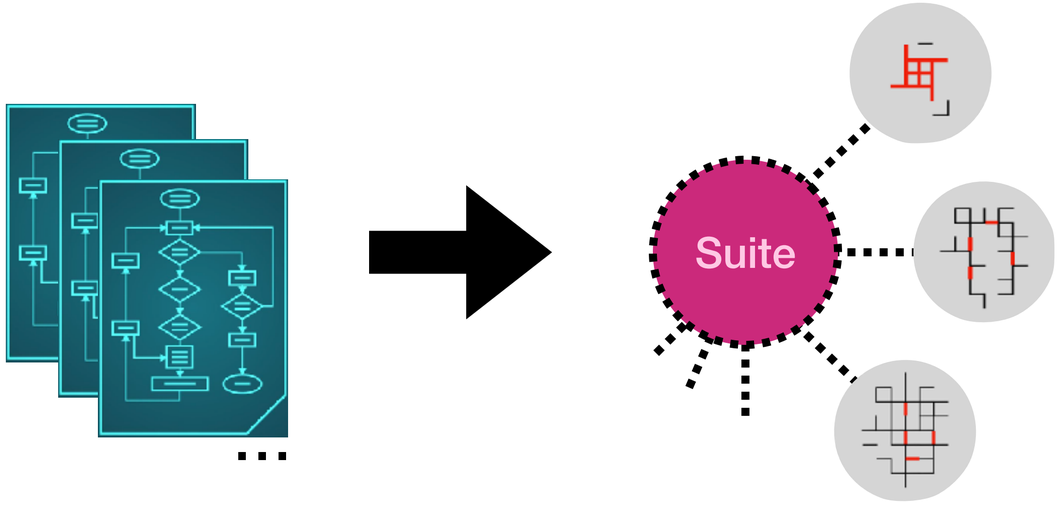


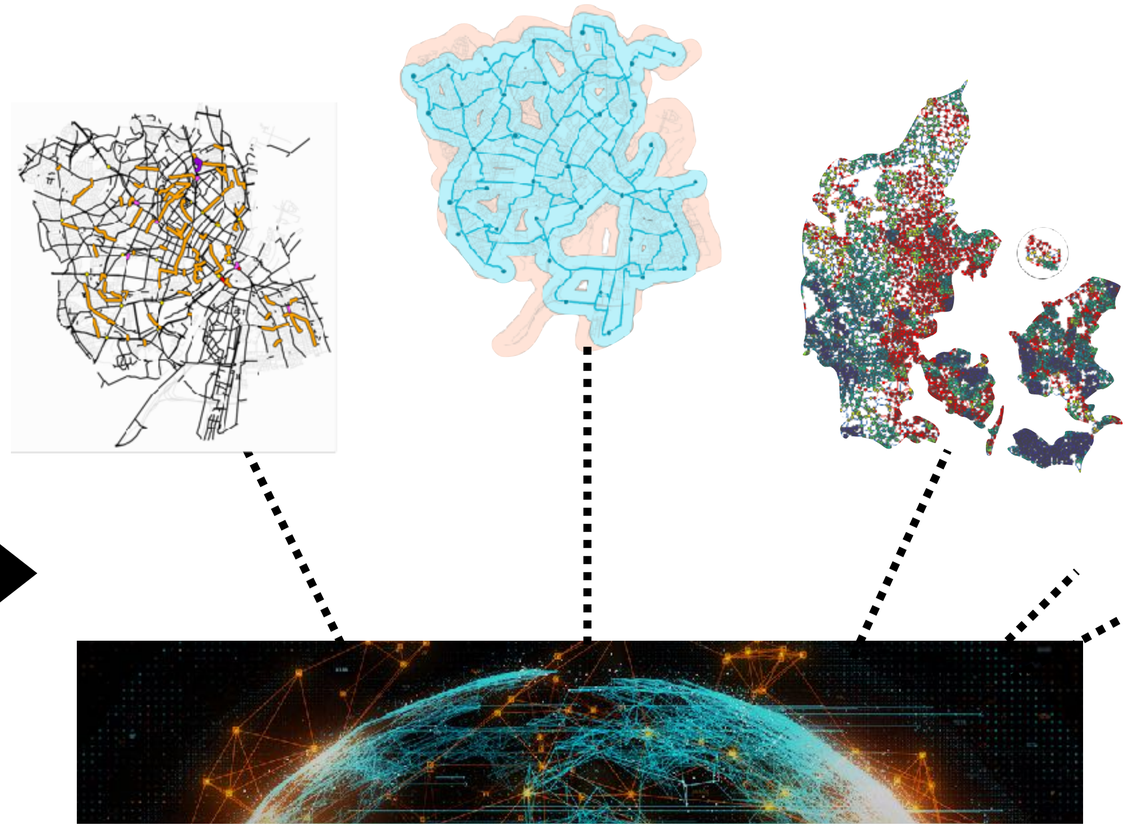
1) Research software engineer (Manuel) creates packaged Python software

Usable by planners

We got funds to 1) Create usable software, 2) Visualize on the web

Innovation Fund Denmark





1) Research software engineer (Manuel) creates packaged Python software

2) Web developer (Moritz) visualizes our tools on a web platform

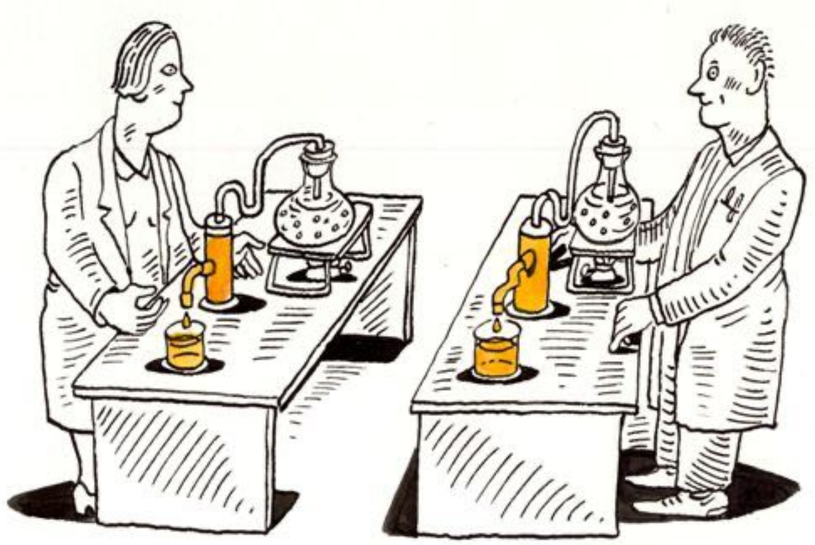
Usable by planners

Guides planners, inspires citizens

Our philosophy: Free, Open-Source, Reproducible

The software should be usable, modifiable, and reproducible by anyone, including proactive citizens, to create a compelling vision for urban cycling in their city.

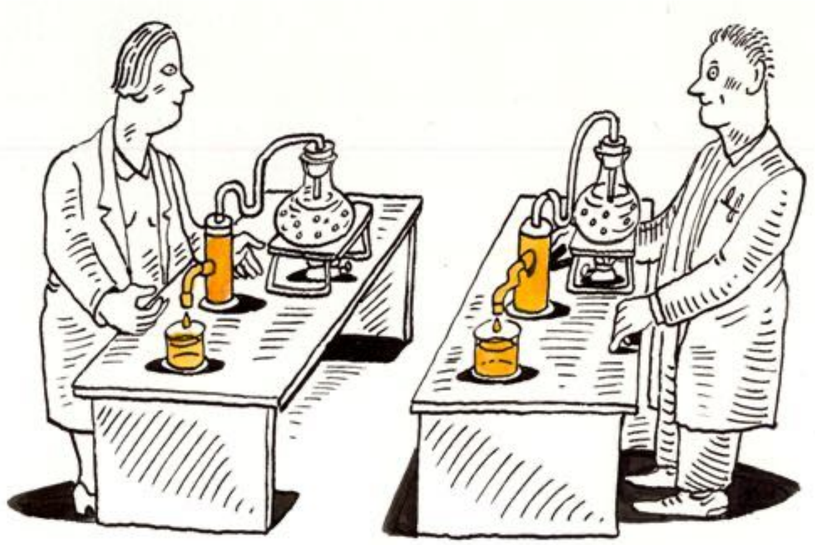




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It should be an alternative to extractive practices
where cities overpay consultants for a one-off report

AGPL v3
Free Software
Free as in Freedom

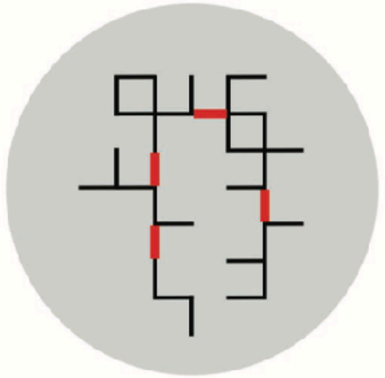
Our tools

We are creating a Python software suite: bikenetkit.org

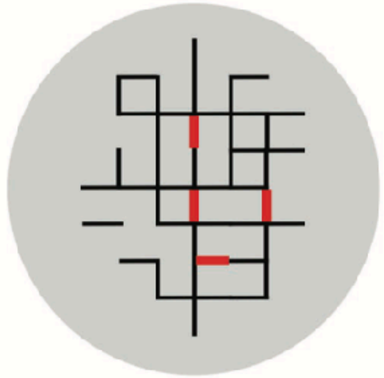
GrowBikeNet



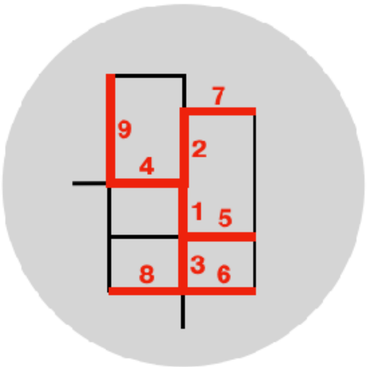
LinkBikeNet



FixBikeNet



SortBikeNet



SUPERBLOCKIFY

Data is based on OpenStreetMap, but any data can be imported

OpenStreetMap has open source, high quality data





Example Turin:
Reprioritizing bike network growth with traffic crash data

https://docs.bikenetkit.org/GrowBikeNet/usage_06_data_imports_for_prioritizing_growth

Data is based on OpenStreetMap, but any data can be imported

OpenStreetMap has open source, high quality data





Example Copenhagen:
Reprioritizing gap identification with citizen survey data

https://docs.bikenetkit.org/FixBikeNet/usage_04_data_imports_for_prioritizing_gaps

How can you help?

- Try out the software. Let us know what works/doesn’t.

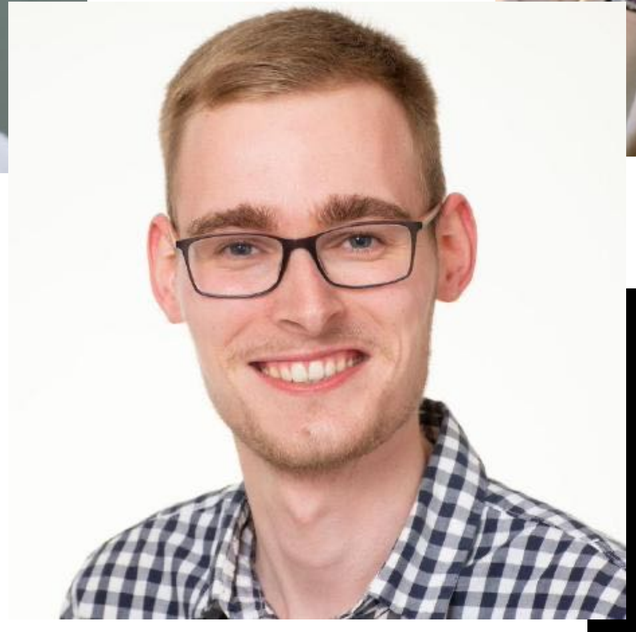
- Follow us / post about it on social media: 

- Star the repositories on Github: https://github.com/BikeNetKit













We launch on Monday, Nov 9th!

bikenetkit.org

Innovation Fund Denmark

JUST STREETS

contact@bikenetkit.org

Bike Net Kit

The digital toolkit for bicycle networks

Free, open-source Python tools for designing
well-connected bicycle infrastructure.

